Product Profitability and Margin Performance Analysis of Nassau Candy Distributor

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Abstract. High sales do not always mean high profit. This study looks at 10,194 order lines of Nassau Candy Distributor (January 2024 to December 2025) to find which products and divisions really earn money and which ones put margins at risk. The overall gross margin is 65.9% on sales of 141,784. Five chocolate bars give 95.1% of all gross profit. Two products with decent sales, Kazookles (7.7% margin) and Lickable Wallpaper (50% margin), earn far less than the chocolate range, and three small sugar products are candidates for discontinuation review. Every product has a fixed price and unit cost, so margins do not move over time and weak margins stay weak until price or cost is changed. The paper gives product-level actions for repricing, cost renegotiation and range review.

1. Introduction

For a distributor, sales volume alone can be misleading. Some products sell a lot but leave little profit, and some use a large share of cost while looking successful. Nassau Candy Distributor sells chocolate, sugar and other candy products made in five factories. The company needs to know which products drive profit, which divisions are weak, and where pricing, sourcing or product range decisions are needed, instead of deciding by intuition.

The study tries to answer four questions:

1. Which products deliver the highest gross profit and gross margin?
2. Are the high-selling products also profitable?
3. How does profitability differ between divisions?
4. Which products are a margin risk?

2. Data and cleaning

The data set has 10,194 rows and 18 columns. Each row is one product line in an order, with sales, units, cost and gross profit. Orders run from 2 January 2024 to 31 December 2025 across four regions. There are 15 products in three divisions (Chocolate, Sugar, Other), made in five factories. The product-to-division and product-to-factory mapping comes from the project brief.

Validation and cleaning steps

- Checked missing values and duplicates: none were found (no duplicate rows or Row IDs).
- Checked Sales, Units and Cost: all values were greater than zero. Gross Profit equalled Sales minus Cost on every row, so no invalid profit records were found and no rows were removed.
- Trimmed text columns and fixed one product name with a spacing error ("Wonka Bar - Scrumdiddlyumptious").
- Each product has exactly one Product ID, so product labels are consistent.

- One label differed from the brief: Fizzy Lifting Drinks is "Sugar" in the data but "Other" in the brief. The brief was followed. It has only 6 orders, so results are not affected.
- Order dates were converted to dates (dd-mm-yyyy). Ship dates fall in 2026-2030, years after the orders, so they look wrong and were not used.

3. Methodology

The analysis was done in Python (pandas, matplotlib, seaborn) and the results are shown in a Streamlit dashboard built with Plotly. Margins for products and divisions are calculated as total profit divided by total sales, not as an average of row margins, so that larger orders get the correct weight.

KPI	Definition
Gross Margin (%)	Gross Profit / Sales x 100
Profit per Unit	Gross Profit / Units
Revenue Contribution	Product sales / total sales
Profit Contribution	Product gross profit / total gross profit
Margin Volatility	Standard deviation of the monthly margin of a product

Products are put into four groups using the median sales (597.50) and median margin (62.31%) of the 15 products. A Pareto analysis finds how many products give 80% of revenue and profit. For cost diagnostics, a product is flagged when its margin is below 55%. The cut-off of 55% was chosen because every chocolate bar is above 64%. A flagged product is marked *Discontinuation review* if its sales are below the median, *Cost renegotiation* if sales are at or above the median and its cost per unit is above the median (1.10), and *Repricing* otherwise.

4. Results

4.1 Overall

Total sales are 141,784, total cost is 48,341 and gross profit is 93,443, which is a gross margin of 65.9%. Sales grew from 57,956 in 2024 to 83,827 in 2025, while the margin stayed almost the same (65.8% and 66.0%).

4.2 Product-level profitability

Product	Division	Sales	Gross profit	Margin %	Profit / unit	Profit share %
Wonka Bar - Scrumdiddlyumptious	Chocolate	27,874.80	19,357.50	69.44	2.50	20.72
Wonka Bar - Triple Dazzle Caramel	Chocolate	28,485.00	18,610.20	65.33	2.45	19.92
Wonka Bar - Milk Chocolate	Chocolate	26,867.75	17,443.37	64.92	2.11	18.67
Wonka Bar - Nutty Crunch Surprise	Chocolate	23,574.95	16,819.95	71.35	2.49	18.00
Wonka Bar - Fudge Mallows	Chocolate	24,890.40	16,593.60	66.67	2.40	17.76
Lickable Wallpaper	Other	7,860.00	3,930.00	50.00	10.00	4.21
Wonka Gum	Other	597.50	310.70	52.00	0.65	0.33
Everlasting Gobstopper	Sugar	130.00	104.00	80.00	8.00	0.11
Kazookles	Other	1,205.75	92.75	7.69	0.25	0.10
Hair Toffee	Sugar	76.50	59.50	77.78	3.50	0.06
Fizzy Lifting Drinks	Other	78.75	47.25	60.00	2.25	0.05
Laffy Taffy	Sugar	53.73	33.48	62.31	1.24	0.04
SweeTARTS	Sugar	61.50	28.70	46.67	0.70	0.03
Nerds	Sugar	15.00	7.00	46.67	0.70	0.01
Fun Dip	Sugar	12.00	4.80	40.00	0.60	0.01

Table 1. Products ranked by gross profit.

The five Wonka chocolate bars are the top five products by profit. Scrumdiddlyumptious earns the most (19,357.50, 20.7% of profit), and Nutty Crunch Surprise has the highest margin among them (71.3%). Everlasting Gobstopper has the best margin overall (80%) but earns only 104 in profit because it sells in very small numbers.

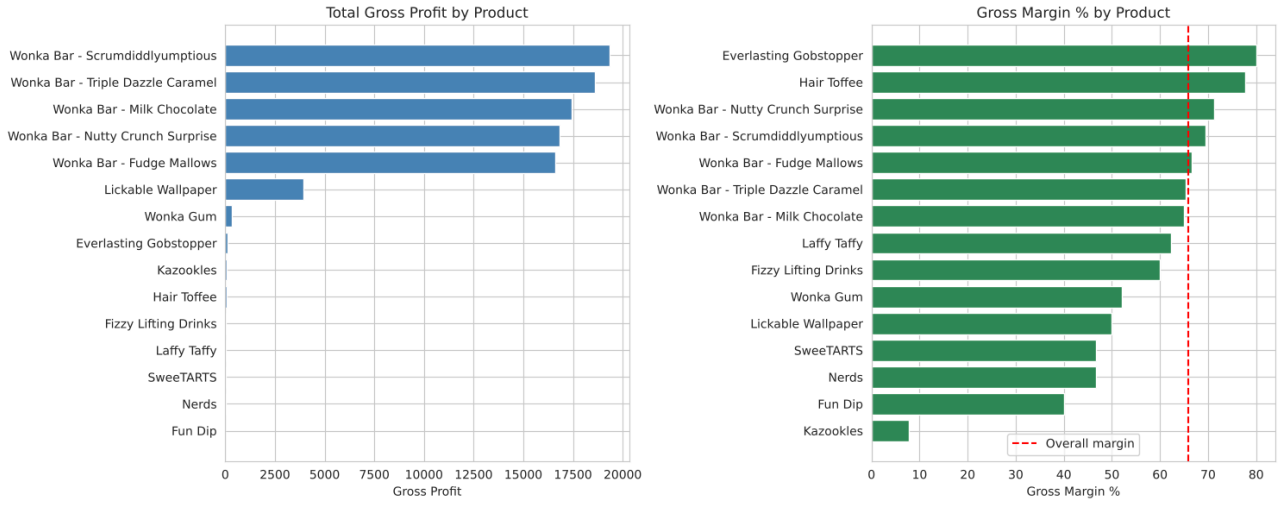


Figure 1. Gross profit and gross margin % by product (red line: overall margin).

4.3 Are high-sales products profitable?

Mostly yes for chocolate, but not for every product. Using the median cut-offs, the products fall into these groups:

- **High sales / high margin:** all five chocolate bars.

- **High sales / low margin:** Lickable Wallpaper (50.0%), Wonka Gum (52.0%) and Kazookles (7.7%).
- **Low sales / high margin:** Everlasting Gobstopper, Hair Toffee and Laffy Taffy.
- **Low sales / low margin:** Fizzy Lifting Drinks, SweeTARTS, Nerds and Fun Dip.

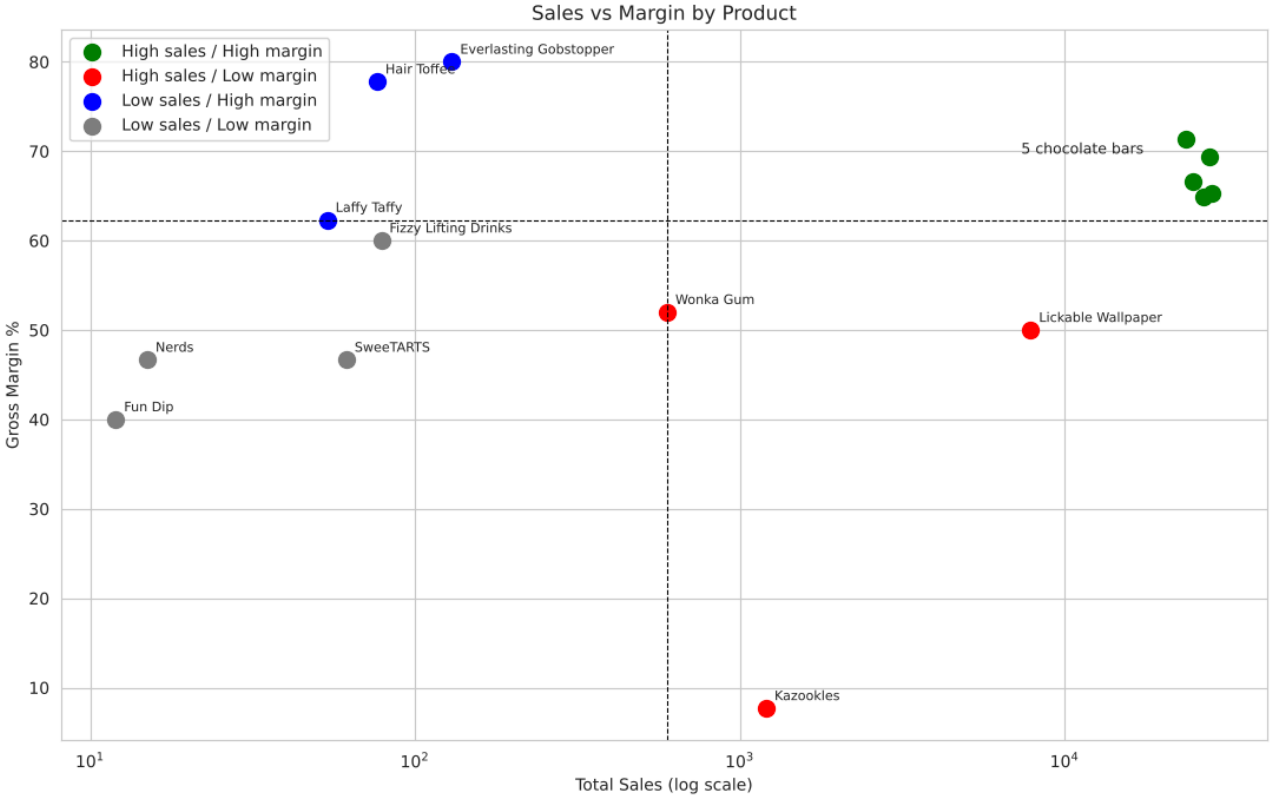


Figure 2. Sales vs margin by product (log scale on sales). Dashed lines are the medians.

Kazookles is the clearest warning. It has the 7th highest sales (1,205.75), but it is sold at 3.25 per unit with a cost of 3.00, so it earns only 0.25 per unit and 0.10% of total profit. Lickable Wallpaper sells well (7,860, 5.5% of revenue) but has the highest unit cost (10.00) on a 20.00 price.

4.4 Margin volatility

Margin volatility is 0 for every product. Each product is sold at a single price with a single unit cost throughout the two years, so margins do not change over time. The differences in margin come from which product is sold, not from discounts or cost changes. This also means a low-margin product will stay low-margin unless its price or cost is changed.

4.5 Division-level performance

Division	Products	Sales	Gross profit	Margin %	Revenue share %	Profit share %
Chocolate	5	131,692.90	88,824.62	67.45	92.88	95.06
Other	4	9,742.00	4,380.70	44.97	6.87	4.69
Sugar	6	348.73	237.48	68.10	0.25	0.25

Table 2. Division summary.

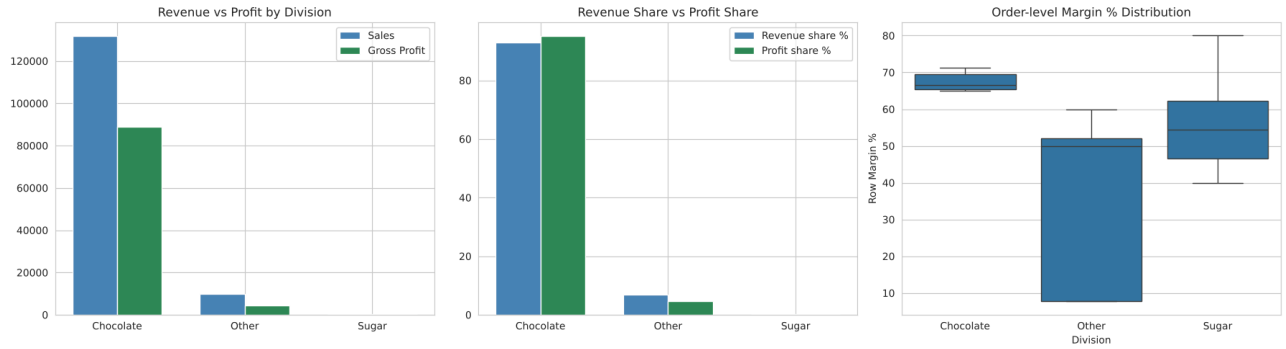


Figure 3. Revenue vs profit, revenue share vs profit share, and order-level margin distribution by division.

Chocolate earns a larger share of profit (95.1%) than of revenue (92.9%), so it is financially efficient. Other earns 4.7% of profit from 6.9% of revenue, a gap of about 2.2 percentage points, and its margin is only 45.0%. Without Kazookles the margin of Other would be 50.2%, so Kazookles is the main cause, but Lickable Wallpaper and Wonka Gum are also below the chocolate level. Sugar has a good margin (68.1%), but it is only 0.25% of revenue, with 3 to 10 orders per product, so its results are not reliable.

By factory, Lot's O' Nuts (69.1% margin) and Wicked Choccy's (65.1%) produce 95% of profit. The Other Factory, which makes Hair Toffee and Kazookles, has a margin of 11.9% because of Kazookles.

4.6 Profit concentration

Five of the 15 products (33%) are enough to reach 80% of revenue, and the same five reach 80% of profit. The top three products give 59.3% of profit and the top five give 95.1%.

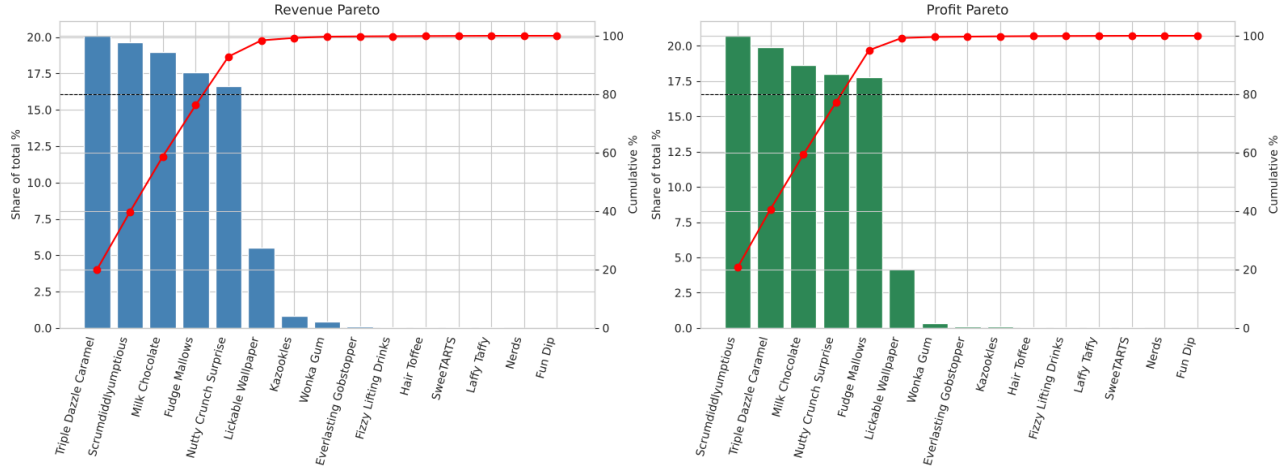


Figure 4. Pareto charts for revenue and profit (dashed line: 80%).

This is a clear dependency risk. The business is mainly a chocolate-bar business, so a supply, price or demand problem in this range would affect almost all of the profit.

4.7 Cost structure diagnostics

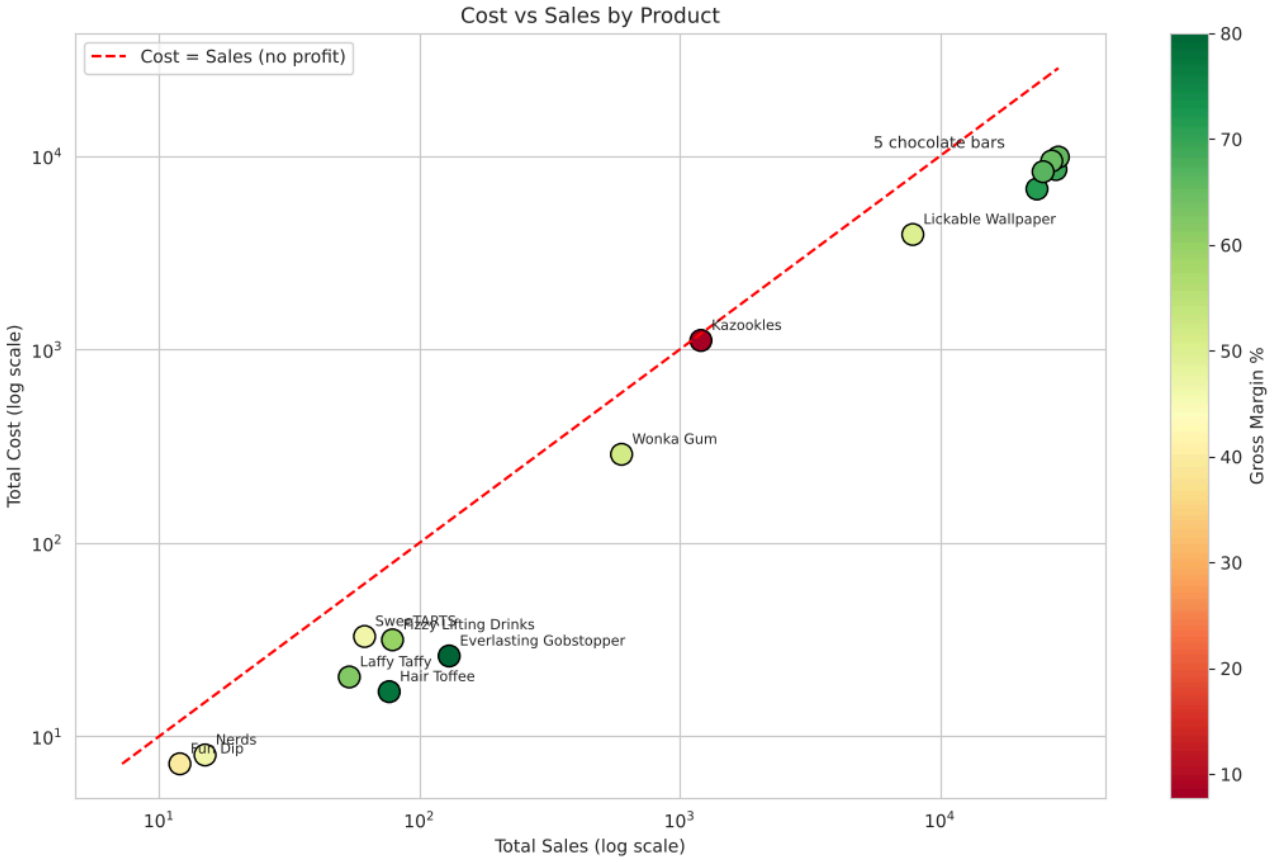


Figure 5. Total cost vs total sales by product (log scales). Products near the dashed line earn very little profit.

Product	Price / unit	Cost / unit	Margin %	Flag
Kazookles	3.25	3.00	7.69	Cost renegotiation
Fun Dip	1.50	0.90	40.00	Discontinuation review
Nerds	1.50	0.80	46.67	Discontinuation review
SweeTARTS	1.50	0.80	46.67	Discontinuation review
Lickable Wallpaper	20.00	10.00	50.00	Cost renegotiation
Wonka Gum	1.25	0.60	52.00	Repricing

Table 3. Products flagged with a margin below 55%. The other 9 products are not flagged.

Together the flagged products make up 6.9% of revenue and 4.7% of profit.

5. Recommendations

1. **Kazookles: renegotiate cost first, then reprice.** With a unit cost of 3.00 and a price of 3.25 there is almost no room. For example, a price of 4.25 would lift the margin to about 29%. If neither cost nor price can be changed, the product should be reviewed for removal.
2. **Lickable Wallpaper: renegotiate cost.** It is a real seller (5.5% of revenue) and worth keeping, but its unit cost of 10.00 on a 20.00 price is high.
3. **Wonka Gum: reprice.** The unit cost is low (0.60), but the price of 1.25 leaves a 52% margin. A price of about 1.70 would give a margin close to the chocolate bars (around 65%).

4. **Fun Dip, Nerds and SweeTARTS: discontinuation review.** They have low sales and margins below 50%, and together they make about 0.04% of profit. Removing them would hardly change profit and would simplify the range.
5. **Protect the chocolate range and reduce dependency.** It earns 95% of profit. Any problem with Lot's O' Nuts or Wicked Choccy's would hit the whole company. Promoting the high-margin products that sell very little (Everlasting Gobstopper, Hair Toffee) is one way to spread the risk.

6. Limitations

- Margin is identical on every order of a product, so the effect of discounts or cost changes cannot be studied with this data.
- Many Sugar products have only 3 to 10 orders, so conclusions about them are weak.
- The 55% margin cut-off and the median-based groups are choices made by the author. The flags change if the cut-off is changed (the dashboard has a slider for this).
- Only gross profit is used. Other costs (shipping, marketing, overheads) are not in the data.
- Ship dates are not reliable, so no delivery or shipping analysis was done.

7. Conclusion

Nassau Candy earns a healthy 65.9% gross margin, but almost all of it comes from five chocolate bars. Sales volume alone would hide the real problems: Kazookles sells reasonably but earns almost nothing, and Lickable Wallpaper and Wonka Gum earn clearly less than the chocolate range. Fixing prices and costs on these three products, reviewing three very small sugar products, and reducing the dependency on the chocolate range are the most useful next steps. The notebook and the Streamlit dashboard allow these results to be reproduced and explored with different date ranges, divisions and margin thresholds.